

NeRF and Gaussian-Based Relighting Advances

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Abstract

Neural Radiance Fields (NeRF) have emerged as a groundbreaking framework for **novel view synthesis**, enabling the generation of highly realistic 3D reconstructions from 2D images. However, **NeRF struggles with relighting**, as it encodes both geometry and illumination into a single implicit representation, preventing the modification of scene lighting. This limitation has driven extensive research into **relightable NeRF variants**, focusing on disentangling material properties, surface normals, and environmental illumination.

The introduction of **3D Gaussian Splatting (3DGS)** has revolutionized neural scene representations by providing an **efficient and real-time** alternative to NeRF. Unlike NeRF, which relies on implicit multi-layer perceptrons (MLPs), **Gaussian splatting models scenes as explicit point-based primitives**, enabling faster rendering and better support for **physically-based relighting**. This shift has enabled advanced relighting techniques that leverage **BRDF-based material estimation**, **environment illumination modeling**, and **differentiable light transport simulation**.

The motivation behind this survey is to explore the **evolution of relighting techniques** in neural rendering, comparing how different approaches tackle **illumination estimation**, **material decomposition**, and **global light transport**. Our objective is to analyze key advancements in **relightable NeRF** and **Gaussian splatting-based methods**, studying how they have progressively improved in **realism**, **efficiency**, and **robustness**.

In this survey, we systematically examine the **timeline of relighting research**, focusing on major contributions from 2020 to 2025. We compare **each relighting approach** in terms of its **methodology**, **improvements over predecessors**, and **practical applications**. Key topics covered include **BRDF decomposition**, **inverse rendering**, **differentiable light transport**, **global illumination modeling**, and **Gaussian-based relighting frameworks**. This analysis provides a comprehensive understanding of how neural scene representations are transitioning from **static appearance models** to **fully editable and relightable scene representations**, bridging the gap between traditional physically-based rendering and neural rendering.

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1 Introduction

NeRF has fundamentally transformed novel view synthesis, but its limitation in **relighting** motivated researchers to extend its capabilities. Over the years, multiple papers introduced

methods to **factorize lighting and material properties** for relightable NeRF models. Recent approaches leverage **Gaussian splatting**, enabling efficient scene decomposition.

2 Methodology

The report follows a chronological structure, summarizing key relighting papers from **2020 to 2025**. Each paper is explained in detail, focusing on **high-level insights**, without mathematical formulations. Papers are compared against their predecessors, highlighting significant advancements in **relighting accuracy**, **material separation**, **lighting manipulation**, and **computational efficiency**.

3 Timeline of Research in Relightable NeRF and Gaussian Splatting

3.1 2021: Initial Steps Towards Relightable NeRF

NeRFactor [1] introduced factorization of NeRF into surface-based components, enabling estimation of albedo, normal maps, and BRDF parameters. This method allowed for relighting under novel illumination conditions but suffered from limited indirect illumination modeling, restricting realistic shadow and global lighting effects.

Neural Light Transport [2] relighting and novel view synthesis by learning a 6D light transport function (surface position (UV) + incident light direction (2D) + viewing direction (2D)). It takes images in an OLAT manner. NLT separates diffuse base rendering and non-diffuse residuals. Diffuse rendering that captures only direct illumination and hard shadows based on simple Lambertian shading. It captures indirect illumination by learning residual corrections for non-diffuse and global light transport effects.

NeRV [3] differs from other NeRF-based relighting research by introducing Neural Visibility Fields (NVFs), which efficiently approximate scene visibility and indirect illumination, making it significantly more scalable to complex lighting conditions. Unlike earlier methods like NeRFactor and Neural Light Transport that struggled with global illumination due to computational costs, NeRV trains a neural network to approximate visibility, enabling realistic relighting under arbitrary lighting.

3.2 2022: Improving Global Illumination and Outdoor Relighting

NeRF-OSR [4] was the first NeRF-based approach tailored for outdoor lighting. Separate albedo and illumination, enabling realistic manipulation of sunlight and shadows. Compared to NeRFactor, it handled dynamic lighting conditions, but still struggled with complex material reflectance. Due to the use of second-order SH for lighting, high-frequency indirect illumination is not fully captured. Limited to Lambertian reflectance model, doesn't fully handle highly dynamic lighting conditions.

NeILF [5] introduced a 5D neural incident light field for improved material and illumination estimation. Unlike earlier works that relied on precomputed lighting, NeILF could model inter-reflections and occlusions without requiring multi-bounce ray tracing, pushing neural relighting closer to physically-based rendering.

3.3 2023: Advanced NeRF-Based Relighting And The Emergence of Gaussian Splatting

GS-IR [6] extended GS for inverse rendering by incorporating material estimation. This allowed normal, albedo, roughness, and metallic properties to be stored per Gaussian, bridging the gap between NeRF’s material separation and Gaussian Splatting’s efficiency. It models indirect illumination by handling occlusions through a baking strategy, capturing subtle shadows and realistic bounced lighting by encoding SH.

GaussianShader [7] marked the beginning of relightable Gaussian Splatting by introducing BRDF-based shading functions. Unlike NeRF approaches that relied on ray marching, GaussianShader utilized point-based representations for efficient relighting. Introduces a term known as residual color, represented by SH, to handle indirect illumination.

ReNeRF [8] improved intrinsic decomposition using OLAT-based supervision, making relighting more accurate. Unlike NeRF-OSR, it captured nearfield lighting effects, enabling realistic rendering of specular highlights and shadowing. The learning is guided by a neural architecture that conditions radiance predictions not only on the 3D location and viewing direction but also on the direction and distance to each light source, enabling the model to simulate both direct and complex indirect illumination effects.

IBL-NeRF [9] introduced a novel neural representation called prefiltered radiance fields to efficiently model spatially varying illumination in complex indoor scenes such as reflections, diffuse inter-reflections, and complex occlusions without explicitly tracing multiple bounces of rays. Unlike previous methods (NeRFactor, NeRV) that require Monte Carlo sampling and environment map approximations, IBL-NeRF achieves superior results through efficient image-based lighting approximation and implicit neural representations. Implicitly handles indirect illumination (such as color bleeding, soft shadows, and reflections) without costly computations.

NeILF++ [10] builds upon its predecessor NeILF by significantly improving the modeling of complex illumination and inter-reflections through a unified neural incident light field representation. It explicitly captures detailed interactions between surfaces, enabling precise separation of material properties from intricate lighting. Unlike earlier approaches, NeILF++ incorporates a highly effective mechanism for estimating spatially varying illumination from multi-view images, resulting in more accurate relighting of scenes with complex global illumination and detailed material properties.

LLNeRF [11] introduces an unsupervised method for relighting low-light scenes by decomposing the NeRF into view-dependent (lighting) and view-independent (intrinsic color) components. It enhances illumination, corrects colors, and reduces noise directly within the NeRF optimization process, enabling high-quality novel view synthesis from low-light sRGB images without ground truth supervision.

ReLight My NeRF [12] introduces the ReNe dataset, the first publicly available real-world dataset specifically for joint novel view synthesis and relighting of complex objects under diverse illumination conditions. Unlike prior works that primarily relied on synthetic scenes or limited setups with minimal shadows, this dataset utilizes a sophisticated dual-robotic-arm system for precise camera and lighting pose control, enabling comprehensive modeling of challenging shadows and intricate lighting effects.

3.4 2024: Real-Time Relightable Gaussian Splatting

Relightable 3D Gaussians [13] Differentiable rendering pipeline extending 3D GS by embedding per-point normals, BRDF parameters, and indirect illumination modeled using SH. Introduced BVH-based point ray tracing for occlusion handling. Unlike GaussianShader, this approach improved accuracy in shadow computation and material estimation. Optimization is guided by regularization on normal, depth, and smoothness. This per-point indirect illumination captures spatially varying inter-reflections, significantly improving realism, especially in complex scenes where indirect light varies notably across surfaces. It achieves more accurate lighting decomposition and realistic shading effects compared to methods that treat indirect lighting uniformly or globally.

BiGS [14] separates diffuse, direct, directional scattering, and indirect transport components for each Gaussian. Uses spherical harmonics based light transport modeling to efficiently represent direct and indirect lighting. Introduces BSH to model light-dependent appearance functions that capture both surface and volumetric materials. OLAT datasets to train an inverse rendering framework that captures complex material-light interactions. Outperforms GaussianShader in preserving subsurface scattering and fur details. Currently limited by low-frequency approximations (difficult to represent sharp shadows or extremely detailed reflections).

LumiGauss [15] brought GS to outdoor relighting by incorporating 2D Gaussian representations for inverse rendering. It enhances indirect illumination handling with the use of precomputed radiance transfer, which allows for efficient and realistic rendering of global illumination effects such as soft shadows and interreflections. Difficulty handling strongly reflective surfaces (like glass), might incorrectly learn shadows as part of the albedo in some cases.

3iGS [16] factorized tensorial illumination model for 3DGS to improve viewdependent effects, material estimation, and relighting. Gaussian-specific BRDF representation to capture specular highlights and roughness variations. Unlike GaussianShader, which relies on precomputed cube maps, 3iGS learns a dynamic illumination field to adapt to complex light interactions. Unlike NeRF-based inverse rendering, which requires ray tracing, 3iGS achieves similar relighting quality with significantly lower computation

Ref- Gaussian [17] represents scenes with sets of 2D Gaussian primitives, pixel-level BRDF properties(Disney BRDF model), Gaussian-grounded inter reflection, a novel technique that uses efficient ray-tracing on an extracted mesh to compute indirect lighting. Visibility is computed via ray tracing on meshes extracted using TSDF fusion. Uses split-sum approximation to efficiently handle the rendering equation.

GI-GS [18] models direct diffuse illumination and direct specular illumination using pre-filtered environment maps and split-sum approximation. It also calculates indirect diffuse illumination and ambient occlusion through an efficient path tracing method based on deferred shading. It extends this to world-space illumination using cubemaps to capture global geometry effects. It does not handle indirect specular illumination such as reflections of reflections due to computational costs. It assumes a globally uniform environment map, not addressing spatially varying environment illumination.

ReCap [19] uses cross-environment captures as multi-task supervision, optimization friendly shading function based on a variant of the split-sum approximation (optimized Disney BRDF) and applies physically appropriate post-processing to maintain a linear HDR lighting space. It tackles the albedo-lighting ambiguity, which occurs when methods can't distinguish whether

changes in an object’s appearance are due to the object’s albedo or the environment’s lighting. ReCap outperforms prior methods such as GShader, 3DGS-DR, GS-IR, and R3DG in achieving more realistic relighting results across both diffuse and specular surfaces. They do not explicitly handle indirect illumination or subsurface scattering.

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